MIL-F-45598
30 July 71
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

FEEDER, DELINKING, AIRCRAFT MACHINE GUN: MAU-56/A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of machinegun driven feed mechanism for ammunition belted with M13 links for use on the 7.62mm, M134 (GAU-2B/A), machinegun. The feeder pulls in the link-belted ammunition, strips the ammunition from the links, ejects the empty links, and feeds the delinked ammunition into the machinegun. The feeder includes an electrically operated mechanism for clearing ammunition from the feeder sprocket prior to the transfer of ammunition to the machinegun when the machinegun trigger is released.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-L-58	-	Lacquer; Spraying, Clear and Pigmented Cellulose Nitrate, (General Interior Use).

Military

MIL-P-116	-	Preservation, Methods of.
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.
MIL-L-46150	-	Lubricant, Semi-Fluid: Low Friction (Automatic Weapons).

FSC 1005

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STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

DRAWINGS

U.S. Army Weapons Command

F11701120	-	Feeder, Delinking, Aircraft Machine Gun, MAU-56/A.

(Product drawings referenced in this specification form a part of F11701120.)

PUBLICATIONS

U.S. Army Weapons Command

P11701120	-	Packaging Data Sheet for Feeder, Delinking, Aircraft Machine Gun, Mau-56/A.
IEL 11701120	-	Index of Inspection Equipment Lists.

(Acceptance inspection equipment drawings referenced in this specification form a part of IEL 11701120.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of the first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

3.2 Materials and construction. Feeders and parts shall conform to the materials and construction requirements specified herein, on Drawing F11701120 and the drawings applicable thereto, and shall be in accordance with the applicable materials and construction provisions of MIL-W-13855.

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3.3 Design. Feeders and parts shall conform to the design specified herein, on Drawing F11701120 and drawings applicable thereto, and shall be in accordance with the applicable design provisions of MIL-W-13855.

3.3.1 Cam guide. The cam guide shall be securely retained in the feeder housing by the screw so that there shall be no relative movement. The cam guide shall guide the cartridges that have been stripped from the links by the push rods to the feeder sprocket when the feeder is operated.

3.3.2 Clearing guide and solenoid group.

3.3.2.1 Clearing guide. The clearing guide shall rotate through its full range of travel between the two spring pins on the feeder housing, without binding.

3.3.2.2 Clearing guide spring. The clearing guide spring shall be securely retained on the feeder housing by the screws and shall retain the clearing guide in position in the slots of the feeder housing.

3.3.2.3 Link and pivot arm. The link and pivot arm shall be attached to the solenoid plunger, feeder housing, and clearing guide by the spring pins and when the solenoid plunger is actuated, the link and pivot arm shall rotate on the spring pins in the clearing guide and feeder housing without binding.

3.3.2.4 Solenoid. The solenoid shall be securely attached to the feeder housing by the screws so that there shall be no relative movement. The solenoid plunger shall move through its full range of travel without binding and shall be returned to and held in the extended position by spring action. Depression of the solenoid plunger shall cause rotational movement of the clearing guide, by action of the link and pivot arm, moving the clearing guide into the feed position. Releasing the solenoid plunger shall allow the clearing guide to rotate into the clearing position.

3.3.3 Entrance guide, link guide and nose guide. The entrance guide, link guide, and nose guide shall be securely attached to the feeder housing by the screws. There shall be no relative movement of these parts.

3.3.4 Feeder cam housing and insert. The feeder cam housing shall be securely attached to the feeder housing by the screws and the insert shall be securely attached to the feeder cam housing by the screws. There shall be no relative movement of these parts.

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3.3.5 Feeder guide. The feeder guide shall be securely retained on the feeder housing by the screws so that there shall be no relative movement.

3.3.6 Feeder shaft group. The feeder gear, stripper sleeve and pin assembly, push rod guide assembly, and feeder sprocket shall be securely retained on the feeder shaft by the spring pins. The spring pin retaining the feeder gear to the feeder shaft shall be retained in the feeder gear by the staked pin spring as specified on the applicable drawing. There shall be no relative movement of these parts on the feeder shaft. The feeder shaft, with the attached parts, shall rotate in the bearings in the feeder cam housing and the forward guide plate without binding.

3.3.7 Forward guide and bearing assembly plate. The forward guide and bearing assembly plate shall be securely retained on the feeder housing by the screws so that there shall be no relative movement. The flanged bearing shall be a press fit in the forward guide plate so that there shall be no relative movement of the bearing in the plate.

3.3.8 Identification plate. The identification plate shall be securely attached to the feeder housing by the drive screws so that there shall be no relative movement.

3.3.9 Push rod assemblies and push rod rollers. The push rod assemblies shall move through their full range of travel in the feeder cam housing, push rod guide assembly, and feeder gear without binding when the feeder gear and feeder shaft are rotated. The push rod pins shall be a press fit in the push rods and the push rod rollers shall rotate on the pins without binding. The push rod assemblies shall strip cartridges from the links and position the stripped cartridges into the stripper sleeve and pin assembly and the cam guide when the feeder is operated.

3.3.10 Round direction plate. The round direction plate shall be securely attached to the feeder housing by adhesive stated on B11701125 so that there shall be no relative movement.

3.3.11 Timing pin and button. The button shall be securely retained on the timing pin by the spring pin. The timing pin shall move through its full range of travel without binding and shall be returned to and held in the extended position by action of the spring.

3.3.12 Lubrication. The push rods and the gear teeth shall be coated with lubricating oil conforming to MIL-L-46150. All other surfaces shall be coated with preservative oil specified on the applicable drawings. The ball bearing shall be protected against contamination by cleaning solvent during cleaning operations.

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3.3.13 Timing. With the timing pin engaged in each of the seven timing positions, the dimensional relationship of the feeder sprocket teeth to the feeder gear shall be within the limits specified on the applicable drawings.

3.3.14 Torque. The torque required to manually start rotation of the feeder shaft without ammunition shall be determined for each feeder.

3.3.15 Solenoid dielectric withstanding voltage. Solenoid shall withstand 590 to 600 volts root mean square at a frequency of 60 Hertz (Hz) for one second without disruptive discharge or deterioration. Testing shall be as specified in 4.5.3.1 and 4.6.1.

3.3.16 Solenoid operation. The solenoid shall actuate the clearing guide and position the clearing guide into the feed position when 25 volts direct current (vdc) is applied to the solenoid and when the solenoid is de-energized it shall allow the clearing guide to return to the clearing position under spring action of the solenoid plunger. Testing shall be as specified in 4.5.3.1 and 4.6.2.

3.3.17 Functioning. The feeder shall operate without stoppage or breakage of parts attributable to the feeder. Testing shall be as specified in 4.5.3.1 and 4.6.3. The feeder shall completely strip all rounds from the links, shall feed the stripped rounds into the machinegun or test equipment simulating the machinegun, shall eject the empty links from the feeder, and perform its clearing function. Links may remain in the exit chute after firing.

3.3.18 Interchangeability. Unless otherwise specified on the drawings, all parts shall be interchangeable. (In normal assembly operations there shall be no objections interposed to preferential assembly of parts provided that all parts are dimensionally acceptable.) Feeders and repair parts shall be capable of meeting the interchangeability tests specified in 4.5.3.2 and 4.6.4.

3.3.19 Endurance. Feeders shall be capable of withstanding an endurance test of 30,000 rounds with at least 200 clearing operations without functioning failures (see 3.3.17), or no more than one clearing malfunction attributable to the feeder and the replacement of shear pins as a result of a malfunction. Testing shall be in accordance with 4.5.3.3 and 4.6.5.

3.4 Marking. Each feeder shall be marked in accordance with the applicable drawings and MIL-W-13855.

3.5 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. First article samples shall be selected from early production and submitted for acceptance in accordance with contract requirements. The samples shall be representative of the production processes to be used during quantity production. The first article samples shall be subjected to the quality conformance inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract (see 6.1).

4.5 Quality conformance inspection.

4.5.1 Inspection lot. The formation, size, and presentation of inspection lots of feeders, parts, and packaging shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions, and contractor's delivery schedule, and within the limitations of MIL-W-13855. Feeders shall be assembled from lots of component parts that have met all inspection requirements specified herein. Interchangeability and endurance test lot size shall be as specified in 4.5.3.2.1.1 and 4.5.3.3.1.

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4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of component parts and concurrent repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1). The contractor's examination of these parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

4.5.2.2 Feeders. Final examination of each feeder shall be performed after completion of all testing and just prior to preservation and packaging. Each feeder shall be examined as specified below. Feeders failing to meet the requirements shall be rejected. Each step in the examination shall include a visual examination for proper cleaning and presence of the specified protective coating and to determine the general quality, completeness of manufacture, assembly, and workmanship.

4.5.2.2.1 Feeder cam housing and insert. Manually examine the feeder cam housing and the insert to determine compliance with 3.3.4.

4.5.2.2.2 Entrance guide, link guide, and nose guide. Manually examine the entrance guide, link guide, and nose guide to determine compliance with 3.3.3.

4.5.2.2.3 Clearing guide and solenoid group.

a. Manually examine the clearing guide, clearing guide spring, link, pivot arm, and solenoid to determine compliance with 3.3.2.1 through 3.3.2.4.
b. Visually examine the fingers on the clearing guide to assure freedom from burs, cracks, and mutilations.

4.5.2.2.4 Timing pin and button. Manually examine the timing pin and button to determine compliance with 3.3.11.

4.5.2.2.5 Forward guide and bearing assembly plate.

a. Manually examine the forward guide and bearing assembly plate to determine compliance with 3.3.7.
b. Visually examine the cartridge guide on the forward guide plate to assure freedom from burs, cracks, and mutilations.

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4.5.2.2.6 Feeder guide.

a. Manually examine the feeder guide to determine compliance with 3.3.5.
b. Visually examine the feeder guide to assure freedom from burs, cracks, and mutilations.

4.5.2.2.7 Cam guide. Manually examine the cam guide to determine compliance with 3.3.1.

4.5.2.2.8 Feeder shaft group.

a. Manually examine the feeder gear, stripper sleeve and pin assembly, push rod guide assembly, feeder sprocket, and a feeder shaft to determine compliance with 3.3.6.
b. Visually examine the sprocket to assure freedom from burs, cracks, and mutilations.
c. Visually examine the gear teeth to assure freedom from burs, cracks, and mutilations.
d. Visually examine to assure that the spring pin that retains the feeder gear to the feeder shaft is flush or below the surface of the feeder gear (see Drawing F11701120).

4.5.2.2.9 Push rod assemblies and push rod rollers. Manually rotate the feed gear and feeder shaft to assure proper function of the push rod assemblies (see 3.3.9).

4.5.2.2.10 Round direction plate. Visually and manually examine the round direction plate to determine compliance with 3.3.10.

4.5.2.2.11 All markings required by applicable drawings shall be visually examined for clarity and legibility (see 3.4). Visually and manually examine the identification plate to determine compliance with 3.3.8 and to assure that all pertinent data has been applied.

4.5.2.2.12 Lubrication. Visually examine the feeder to assure that all functioning surfaces have been lubricated (see 3.3.12).

4.5.2.2.13 Timing. Feeders shall be examined for the timing requirement (see 3.3.13) using inspection equipment conforming to Drawing F11702138. The feeder shall be assembled to the timing gage and the feeder sprocket shall be rotated until the timing pin can be engaged in one of the feeder gear timing holes. The gear alinement pin of the gage shall then be depressed between the nearest feeder gear teeth as the timing pin is released. With the feeder sprocket in this position for each of the seven timing positions, the gage shall indicate whether the timing requirement has been met.

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4.5.2.2.14 Torque. The torque required to manually start rotation of the feeder shaft without ammunition (see 3.3.14) shall be measured and recorded. The recorded measured values shall be forwarded in accordance with 6.1.2.

4.5.2.2.15 Inspection approval stamp. Feeders which have passed all examinations after successful completion of testing shall be stamped with the Department of Defense complete inspection approval stamp in the area provided on the identification plate. The Government representative will control the stamps used for such stampings to assure successful completion of all required examination and test.

4.5.2.3 Packaging. Examination of packaging of feeders shall be performed in accordance with the classification of defects and acceptable quality levels (AQL's) specified in 4.5.2.3.1 (see 6.1). Sample size shall be in accordance with MIL-STD-105, using inspection level I. The following provisions shall apply:

a. The AQL's are specified as percent defective.
b. An individual AQL is specified for each listed defect, not for a group of defects.
c. Examination for packaging defects specified in 4.5.2.3.1 shall apply to each item of the applicable sample of feeders, interior packages, or exterior containers as applicable.

4.5.2.3.1 Classification of defects for packaging. (Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet P11701120).

Categories	Defect	AQL
Critical:	None defined.	
Major:		
101	Illegible or incorrect marking.	1.0
102	Improper location of marking.	1.0
103	Improper level of packaging and packing (see procurement documents).	1.0
104	Inadequate cleaning and drying.	1.5
105	Missing or inadequate quantity of desiccant.	1.5
106	Humidity indicator missing.	1.5
107	Improper closure of unit packages.	1.5
108	Improper assembly of package: position of assemblies in supports and position of separators.	1.5
109	Improper closure of shipping containers.	1.5
Minor:	None defined.	

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4.5.2.3.2 Examination of packaging of repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1).

4.5.3 Testing.

4.5.3.1 Solenoid dielectric withstanding voltage, solenoid operation, and functioning testing. The contractor shall test each feeder for solenoid dielectric withstanding voltage, solenoid operation, and functioning using the test method specified in 4.6.1, 4.6.2, and 4.6.3 respectively. Feeders failing to meet the requirements shall be rejected.

4.5.3.2 Interchangeability testing.

4.5.3.2.1 In plant.

4.5.3.2.1.1 Feeders. The contractor shall test a sample of three feeders selected by the Government representative, from each month's production for interchangeability using the test method specified in 4.6.4.1.1. Feeders taken for interchangeability testing shall have been found satisfactory in all other examinations and tests. The three feeders shall be inspected for timing and torque before and after interchange of parts using the inspection methods specified in 4.5.2.2.13 and 4.5.2.2.14. In addition, the feeders shall be tested for functioning requirement after interchange of parts using the test method specified in 4.6.3. Failure of the interchangeability test shall cause retest or rejection of the represented lot. At the discretion of the Government representative, an interchangeability retest may be allowed without reconditioning the lot of feeders. Failure in the retest shall cause rejection of the represented lot subject to reconditioning and further test as a reconditioned lot. A sample of six feeders from each retest or reconditioned lot shall be tested using the same procedure described above.

4.5.3.2.1.2 Concurrent repair parts. The contractor shall subject at least two parts from each inspection lot of concurrent repair parts to the interchangeability test specified in 4.6.4.1.2. Failure of any part to meet the requirements shall be cause for rejection of the represented lot of parts subject to reconditioning and further test as a reconditioned lot. A sample of double the number of parts used in the original test shall be tested from each reconditioned lot using the test method specified in 4.6.4.1.2.

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4.5.3.2.2 Interplant. When feeders are manufactured concurrently by more than one contractor, each contractor shall forward monthly to the testing agency specified in the contract, three feeders for the interplant interchangeability test specified in 4.6.4.2 (see 6.1). The contractor will be informed of any failure of the feeders to meet prescribed requirements.

4.5.3.3 Endurance testing.

4.5.3.3.1 Lot size. The initial test lot of feeders shall consist of the first month's production. Thereafter, endurance test lots shall each consist of 100 feeders or a month's production, whichever is larger.

4.5.3.3.2 Procedure. One feeder, selected by the Government representative from each endurance lot shall be tested by the contractor for endurance using the test method specified in 4.6.5. If the endurance requirements are not met, the represented lot shall be rejected subject to retest or reconditioning and further test as a reconditioned lot. An endurance retest of two other feeders from the same lot shall be made without reconditioning the represented lot, unless in the opinion of the Government representative the failure indicates serious defects in the item, in which case retest shall be made only when authorized by the procuring agency. Failure of any feeder in the retest to meet the requirements shall cause rejection of the represented lot subject to reconditioning and further testing as a reconditioned lot. Prior to submission of a lot of feeders as a reconditioned lot, the cause of failure shall be determined and contractor correction shall be effected on all feeders and parts in production. Sample size and test method for reconditioned lots shall be the same as for retest.

4.5.3.4 Certification. Unless otherwise specified, the contractor shall be responsible for testing necessary to accomplish certification. For each inspection lot of feeders, the contractor shall provide the Government representative with certified statements of compliance with applicable drawings and specifications of the following materials and processes specified on Drawing F11701120.

a. Plate, identification (B11701098).
b. Plate, round direction (B11701125).
c. Screw, drive, round head (MS21318-8).
d. Lubricating oil (MIL-L-46150)(see 3.3.12).
e. Lacquer (TT-L-58).
f. Touch-up (MIL-W-13855).

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4.5.3.5 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.1). This will include chemical analysis and physical tests of materials, and tests of protective finish, heat treatment, bonding, and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.3.6 Packaging testing.

4.5.3.6.1 Feeders.

4.5.3.6.1.1 The contractor shall furnish the Government representative with certification that the packaging materials conform to the applicable packaging data sheets and specifications.

4.5.3.6.1.2 Determination of cleanliness. The contractor shall test items selected by the Government representative from each inspection lot for cleanliness using the test methods specified in 4.6.6.1. Sampling shall be in accordance with MIL-P-116.

4.5.3.6.1.3 Heat seal and vacuum retention. The contractor shall test level A unit packages selected by the Government representative from each inspection lot for heat seal and vacuum retention using the test methods specified in 4.6.6.2 and 4.6.6.3, respectively. Sampling shall be in accordance with MIL-P-116.

4.5.3.6.2 Repair parts. Testing of packaging of repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1).

4.5.4 Inspection equipment.

4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified, responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on lists contained on Index of Inspection Equipment List, Drawing 11701120, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.4.2 When specified in the procurement document (see 6.1), copies of drawings of contractor designed inspection equipment shall be forwarded as specified in the contract.

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4.5.4.3 Ammunition and links (see 6.1).

4.5.4.3.1 Ammunition. Unless otherwise specified in procurement documents, Government standard 7.62mm ball cartridges or XM172 (less primer) ball, inert loaded, smooth case cartridges shall be used for testing feeders.

4,5,4,3.2 Links. Government standard 7.62mm, M13 links shall be used for linking ammunition.

4.6 Test methods.

4.6.1 Solenoid dielectric withstanding voltage test. Feeders shall be tested for solenoid dielectric withstanding voltage requirement (see 3.3.15) using a current limited high potential tester and electronic timer conforming to Drawing A7273225 and A7273227 respectively, set up in accordance with Drawing B11701985. The high potential voltage shall be applied from one second between either coil terminal "B" or "E" (see Drawing C11701134) and grounded metal parts. High potential test voltages shall be reduced 20 percent for retest of solenoids previously tested at the voltages specified in 3.3.15.

4.6.2 Solenoid operation test. Feeders shall be tested for solenoid operation requirement (see 3.3.16) after the solenoid dielectric withstanding voltage test using a power supply conforming to Drawing A11701912 set up in accordance with Drawing B11701985 by applying 25 vdc to the solenoid. (Do not energize the solenoid for more than 5 seconds.)

4.6.3 Functioning test.

4.6.3.1 Feeders shall be tested for functioning requirement (see 3.3.17) using the test equipment listed below, set up and operated in accordance with Drawing B11701985.

Test equipment	Drawing number	
Power supply (solenoid)	A11701912	
Power supply (motor)	A11701913	
Oscillograph	A11701914	
Electric drive	D11686350	(GOVERNMENT FURNISHED)
Test stand	F11702136	
Electrical test set	F11702137	
Galvanometer	A11018983	

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4.6.3.2 Two belts of 150 rounds each of linked dummy ammunition shall be used for testing each feeder. The feeder shall be accelerated up to a feeding steady state rate of 4,000 to 4,400 rounds per minute before power is removed from the electric drive and the clearing action is initiated at approximately the 50th and 100th round of each belt. Power to the feeder solenoid shall be applied at the same time power to the electric drive is applied and power shall be removed from the drive 0.135 to 0.165 seconds after the feeder solenoid is de-energized.

4.6.4 Interchange of parts.

4.6.4.1 In plant.

4.6.4.1.1 Feeders. Feeders shall be tested for interchange of parts (see 3.3.18) by disassembling and then reassembling parts using the parts and pre-arranged system specified below. Interchange of parts shall be accomplished by dividing the parts of each feeder into three groups of non-mating parts as shown below and distributing the groups into three different trays until each tray contains parts for a complete feeder. Groups of parts from the first feeder shall be taken in order and placed in trays 1 through 3; groups of parts from the second feeder shall be taken in order and placed in trays 2, 3, and 1; groups of parts from the third feeder shall be taken in order and placed in trays 3, 1, and 2. Commercial parts such as screws, spring pins, etc., shall be placed in the same tray as their mating or associate part. Any commercial part rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test. The feeders shall be reassembled using only those parts which are in the same tray.

Groups of non-mating parts

Group I

Button (11701056)
Gear, feeder (11701131)
Guide, clearing (11701140)
Guide, link (11686380)
Guide, nose (11686381)
Housing, feeder cam (11701139)
Plate, fwd guide & brg assy (11701216)
Solenoid (11701134)

Group II

Rearing, ball (11701116)
Guide assy, push rod (11701218)
Housing, feeder (11686378)
Insert (11701129)
Link (11701127)
Roller, push rod (11701122)
Stripper sleeve & pin assy (11701217)
Sprocket, feeder (11701130)

Group III

Arm, pivot (11701128)
Guide, cam (11701137)
Guide, entrance (11686379)
Guide, feeder (11701132)
Pin, timing (11701055)
Push rod assembly (11701126)
Shaft, feeder (11701136)
Spring, clearing (11701124)

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4.6.4.1.2 Concurrent repair parts. Concurrent repair parts shall be tested for interchangeability requirement (see 3.3.18) by disassembling two feeders, previously tested in 4.5.3.2.1.1, as necessary and then reassembling them using the concurrent repair parts. No hand refinement of parts will be allowed, and the feeders shall operate and function properly. This test may be performed independently of the feeder interchangeability test specified in 4.5.3.2.1.1, and at more frequent intervals using accepted feeders taken from current production.

4.6.4.2 Interplant. Feeders to be subjected to the interplant interchangeability test shall be inspected for timing, torque, and functioning using the inspection methods specified in 4.5.2.2.13, 4.5.2.2.14, and 4.6.3 respectively, to assure proper operation before parts are interchanged. Parts shall then be disassembled from the feeder and identified as to manufacturer. Disassembled parts shall be mixed and the feeders shall be reassembled by random selection without altering or hand fitting any part. After assembly, the feeders shall be inspected for timing, torque, and functioning using the inspection methods specified in 4.5.2.2.13, 4.5.2.2.14, and 4.6.3 respectively. Before feeders are returned to the contractors, the original parts shall be reassembled to their respective feeders.

4.6.5 Endurance test.

4.6.5.1 Feeders shall be tested for endurance (see 3.3.19) using the test equipment specified in 4.6.3.1.

4.6.5.2 Linked ammunition in belts of 150 rounds each (see 4.5.4.3) shall be used. The feeders shall be accelerated up to a feeding steady state rate of 4,000 to 4,400 rounds per minute before the power is removed from the drive and the clearing action is initiated approximately in the middle of each belt. Feeders shall be checked after each 1,500 rounds for correct timing and loosening of parts. All loosened parts shall be tightened and recorded. In addition, the feeders shall be lubricated after each 1,500 rounds and at the end of each day's testing.

4.6.5.3 Endurance tested feeders shall be disposed of as specified in the contract (see 6.1).

4.6.6 Packaging tests.

4.6.6.1 Determination of cleanliness. The applicable surfaces of each sample unit shall be subjected to the determination of cleanliness test specified in MIL-P-116.

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4.6.6.2 Heat seal. The sample level A unit packages shall be subjected to the heat seal test specified in MIL-P-116.

4.6.6.3 Vacuum retention. A sufficient vacuum shall be drawn to cause the flexible barrier of level A unit packages to cling snugly to the enclosed item. Care shall be exercised to insure that an excessive amount of vacuum is not applied which might cause puncture or rupture of the barrier.

4.6.6.3.1 Interpretation of results. After remaining undisturbed at room temperature for two hours, the barrier shall be examined to determine whether it is still taut and retracts against the item when drawn away and quickly released.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. A pilot package shall be forwarded in accordance with the contract (see 6.1). Pilot packages shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of Packaging Data Sheet P11701120.

5.2 Preservation, packaging, packing and marking. Feeders shall be preserved, unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet P11701120 for the level of protection specified (see 6.1).

5.3 Repair parts. Repair parts shall be prepared for delivery in accordance with the applicable packaging data sheets specified in the contract (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement documents.

a. Title, number and date of this specification.
b. List of drawings and specifications pertinent to the feeder, showing applicable revision dates.
c. Requirements for and the number of feeders to be submitted for first article by the contractor (see 3.1 and 4.4).
d. Criteria for examination of component parts and concurrent repair parts (see 4.5.2.1).
e. Criteria for examination of packaging of feeders (see 4.5.2.3) and repair parts (see 4.5.2.3.2).

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f. Criteria for performance of testing of component parts and concurrent repair parts (see 4.5.3.5).
g. Criteria for testing of packaging of repair parts (see 4.5.3.6.2).
h. List of inspection equipment, responsibilities for acquisition, maintenance and disposition of, if other than as specified (see 4.5.4).
i. Place of final inspection and acceptance (see Notes in MIL-W-13855).
j. When required shipping instructions for copies of contractor designed inspection equipment (see 4.5.4.2).
k. Responsibilities for furnishing ammunition and links (see 4.5.4.3).
l. Shipping instructions for feeder and parts when an interplant interchangeability test is required (see 4.5.3.2.2).
m. Disposition instructions of endurance tested feeders and parts (see 4.6.5.3).
n. Procedures and methods for disposition of rejected material.
o. Disposition of Government furnished property.
p. That packages opened for examination shall be repackaged by the contractor at the contractor's expense.
q. Shipping instructions for first article samples and pilot pack (see 3.1 and 5.1).
r. Selection of applicable levels of preservation, packaging and packing (see 5.2).
s. Packaging instructions for repair parts (see 5.3).

6.1.2 Contract data requirement. Monthly reports of the results of final examination, functional testing (see 4.6.3) and torque (see 4.5.2.2.14), shall be specified for delivery on a DD Form 1423 included in the contract.

6.2 When warranted, the contract shall specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Summary List, DD Form 1660.

6.3 Unless otherwise specified (see 6.1.1 h), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Summary List, DD FORM 1660.

6.4 When action by a testing agency is required, work programing will be affected with a testing agency at the earliest practicable date.

6.5 Supersession data. This specification includes the requirements of Springfield Armory Purchase Description SAPD-289A, Amend 2, dated 17 December 1966.

Custodian:
Army - WC
Air Force - 84

Preparing activity:
Army - WC

Project Number:
1005-A419

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-F-45598 Feeder, Delinking, Aircaraft Machine Gun: MAU-56/A

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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